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SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme: **B.E / B.Tech**
Course Code: **MLA02**
Course Name: **FUNDAMENTALS OF MACHINE LEARNING**
Duration: **1:30 Hr**
Max Marks: **50 Marks**
Date : _____

ASSIGNMENT – II

UNIT II – SUPERVISED LEARNING

Topics: *Linear regression and classification, naïve Bayes, discriminant functions, logistic regression, decision trees and pruning, neural networks and back-propagation, SVM, ensembles (bagging, boosting)*

Course Outcome

On successful completion of this assignment, students will be able to:

CO	Course Outcome Statement
CO2	Apply supervised learning algorithms for regression, classification, and predictive modeling tasks to solve structured data problems in practical applications.

Answer all the Tasks

Task 1. The following data relate the number of training epochs x to a validation score y.

x	1	2	3	4	5
y	2.1	3.9	6.2	7.8	10.1

(a) Fit $y = w_0 + w_1x$ by ordinary least squares using the normal equations. (b) Predict y at x = 6. (c) Compute the residual sum of squares, MSE and R^2. (d) Analyse the reliability of predicting at x = 6.

(5) (CO2) (BL4)

Task 2. For mean-centred data with $Sxx = \Sigma x^2 = 10$ and $Sxy = \Sigma xy = 19.9$ (single feature, no intercept needed), compute the ridge-regression slope for $\lambda = 5$. Calculate the percentage shrinkage relative to the least-squares slope and explain the bias–variance effect of increasing λ.

(2) (CO2) (BL4)

Task 3. An e-mail filter uses two binary features: Offer (contains the word 'offer') and Link (contains a hyperlink). Ten labelled e-mails are available.

#	Offer	Link	Class
1	Y	Y	Spam
2	Y	N	Spam
3	Y	Y	Spam
4	N	Y	Spam
5	Y	Y	Spam

#	Offer	Link	Class
6	N	N	Ham
7	N	Y	Ham
8	Y	N	Ham
9	N	N	Ham
10	N	N	Ham

(a) Estimate the priors and the class-conditional probabilities without smoothing. (b) Classify an e-mail with Offer = Y, Link = Y and report the posterior P(Spam | x). (c) Repeat (b) with Laplace smoothing (α = 1). (d) Classify an e-mail with Offer = N, Link = N (with smoothing) and analyse the effect of the conditional-independence assumption and of smoothing.

(6) (CO2) (BL4)

Task 4. A logistic-regression classifier $p = \sigma(w_1x_1 + w_2x_2 + b)$ is initialised with $w = (0, 0)$, $b = 0$ and trained on two samples: $x^{(1)} = (1, 2)$ with $y = 1$, and $x^{(2)} = (2, 1)$ with $y = 0$. Perform one batch gradient-descent step with learning rate $\eta = 0.5$ on the mean cross-entropy loss. Report the gradient, the updated parameters, the predicted probabilities and the loss before and after the update, and analyse the result.

(4) (CO2) (BL4)

Task 5. A node contains 10 training samples (6 Yes, 4 No). A candidate attribute A splits it into $A = a_1$ (4 samples: 4 Yes, 0 No) and $A = a_2$ (6 samples: 2 Yes, 4 No). Calculate the entropy of the node, the information gain of the split and the Gini-impurity decrease, and evaluate whether the split is worthwhile.

(3) (CO2) (BL4)

Task 6. In a decision tree trained on 100 samples, an internal node t with a four-leaf subtree T_t makes 3 training errors, whereas collapsing t into a single leaf would make 8 errors. Using cost-complexity pruning with $R_\alpha(T) = R(T) + \alpha|\text{leaves}|$, compute the critical value α at which the subtree and the single leaf have equal cost, and evaluate whether the subtree should be pruned for $\alpha = 0.01$ and $\alpha = 0.02$.

(3) (CO2) (BL5)

Task 7. A network has two inputs, one sigmoid hidden unit and one sigmoid output unit (no biases): $h = \sigma(w_1x_1 + w_2x_2)$, $\hat{y} = \sigma(vh)$, with $w_1 = 0.4$, $w_2 = 0.3$, $v = 0.5$. For the input $x = (1, 2)$ with target $t = 1$ and squared-error loss $E = \frac{1}{2}(t - \hat{y})^2$, perform one forward pass and one back-propagation update with $\eta = 0.5$. Report h, $\hat{y}$, E, the error terms δ, the gradients, the updated weights and the new loss.

(4) (CO2) (BL4)

Task 8. A linear SVM in two dimensions has $w = (0.5, 0.5)$ and $b = -2$, with the support vectors (3, 3) (class +1) and (1, 1) (class −1). (a) Verify that the support vectors lie on the margin hyperplanes and compute the geometric margin width. (b) Classify the points (4, 2) and (1, 2) and compute their distances from the decision boundary. (c) Compute the hinge loss for (1, 2) with true label −1 and for (2, 2) with true label +1, and state which points violate the margin.

(4) (CO2) (BL4)

Task 9. Independent classifiers, each with accuracy 0.70, are combined by majority voting. Calculate the ensemble accuracy for 3 and 5 classifiers, comment on the gain from adding more classifiers, and analyse what happens to this gain if the classifiers' errors are strongly correlated.

(3) (CO2) (BL4)

Task 10. AdaBoost starts with five training samples of equal weight. The first weak learner misclassifies samples 2 and 5. Compute the weighted error, the learner weight α, the un-normalised and normalised weights after the update, and analyse how boosting changes the focus of the next learner.

(4) (CO2) (BL4)

Task 11. Two classes share the covariance matrix $\Sigma = [[2, 0.5],[0.5, 1]]$ and have means $\mu_1 = (1, 2)^T$ and $\mu_2 = (3, 1)^T$. (a) Derive the discriminant direction $w = \Sigma^{-1}(\mu_2 - \mu_1)$ and the decision threshold for equal priors. (b) Classify $x = (2.5, 1)$. (c) Evaluate whether the decision changes if the priors are $P(\omega_1) = 0.8$ and $P(\omega_2) = 0.2$.

(4) (CO2) (BL4)

Task 12. A bank evaluates three classifiers on 1,000 test loans (100 actual defaulters). Default is the positive class.

Model	TP	FN	FP	TN
A	70	30	90	810
B	85	15	180	720
C	60	40	30	870

A missed defaulter (FN) costs ₹50,000 and a wrongly rejected good applicant (FP) costs ₹5,000.

(a) Compute accuracy, precision, recall and F1-score for each model. [3]

(b) Compute the total expected cost of each model. [2]

(c) Critically evaluate which model should be deployed and explain why accuracy and F1 lead to a different choice than cost. [2]

(d) Propose two modelling or threshold-tuning steps to improve the chosen model and one validation safeguard against over-fitting. [1]

(8) (CO2) (BL6)

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ANSWER SCRIPT EVALUATION SHEET

ASSIGNMENT - II

Programme	B.E / B.Tech	**Course Code**	**MLA02**
Course Name	**Fundamentals of Machine learning.**	**Date**	
Duration	1 Hr 30 Min	**Max Marks**	50

Register No.: _____

TASK 1	Max Marks	Marks Awarded
Task 1	5	5
Task 2	2	2
Task 3	6	6
Task 4	4	4
Task 5	3	3
Task 6	3	3
Task 7	4	4
Task 8	4	4
Task 9	3	3
Task 10	4	4
Task 11	4	4
Task 12	8	8
TOTAL	**50**	50

Total in words:

204
Aamentals of
MACHINE LEARNING ASSIGNMENT - 02

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Task 1- linear Regression

given: $x = 1, 2, 3, 4, 5$

$y = 2.1, 3.9, 6.2, 7.8, 10.1$

a) Fit the linear regression equation

$y = w_0 + w_1 x$

$$\begin{bmatrix} 5 & 15 \\ 15 & 55 \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} = \begin{bmatrix} 30.1 \\ 110.2 \end{bmatrix}$$

$w_0 = 0.05$

$w_1 = 1.99$

$\therefore y = 0.05 + 1.99x$

b) Predict y at $x = 6$

$y = 0.05 + 1.99(6)$

$y = 11.99$

c) cal RSS, MSE & R^2

→ RSS = 0.107

→ MSE = RSS/5 = 0.0214

→ R^2 = 0.9973

d) Reliability

→ R^2 is very close to 1

→ The Model fits the given data very well

∴ The prediction is reasonably reliable but not ~~good~~ guaranteed.

Task 2- Ridge Regression

given: $S_{xx} = 10$

$S_{xy} = 19.9$

$\lambda = 5$

$$w = \frac{S_{xy}}{S_{xx} + \lambda}$$

$$w = \frac{19.9}{10 + 5}$$

$w = 1.3267$

$$w_{LS} = \frac{19.9}{10} = 1.99$$

% shrinkage:

$$\frac{1.99 - 1.3267}{1.99} \times 100$$

33.33%

Bias-variance effect

→ Bias increase

→ variance increase

Task 3 - Naive Bayes classification

a) cal priors & conditional prob

Total spam = 5 P(spam) = 3/10 = 0.5

Total Ham = 5 P(Ham) = 5/10 = 0.5

Coditional prob:

Prob	spam	Ham
offer = Y	4/5	1/5
offer = N	1/5	4/5
link = Y	4/5	1/5
link = N	1/5	4/5 ✓

b) classify offer = Y link = Y without smoothing

$$P(spam|Y,Y) = \frac{0.5 \times 0.8 \times 0.8}{(0.5 \times 0.8 \times 0.8) + (0.5 \times 0.2 \times 0.2)}$$

$P(spam|Y,Y) = 0.9412$

c) with laplace smoothing ($\alpha = 1$)

Spam conditional prob:

$$P(Y|spam) = \frac{4+1}{5+2} = \frac{5}{7}$$

Ham conditional prob:

$$P(Y|Ham) = \frac{1+1}{5+2} = \frac{2}{7}$$

$$P(spam|Y,Y) = \frac{25}{29}$$ ✓

0.8621

d) classify offer = N link = N smoothing

$$P(spam|N,N) = \frac{4}{29}$$

0.1379 ✓

∴ Email is Ham.

Task 4 - logistic Regression

given initial weights : $w_1 = 0$ $w_2 = 0$

Bias = 0

learning rate = 0.5

initial prediction: $p = \sigma(0) = 0.5$

gradient calculation: $\frac{\partial L}{\partial w_1} = 0.25$ ✓

$\frac{\partial L}{\partial w_2} = 0.25$

$\frac{\partial L}{\partial b} = 0$

Updated weights:

$w_1 = -0.125$

$w_2 = 0.125$

$b = 0$ ✓

Predicted prob after update

sample 1 = 0.5312

sample 2 = 0.4688 ✓

loss:

Before update = 0.6931

after update = 0.6326. ✓

TASK-5: Decision tree Entropy & Gini.

Given: Total samples = 10

Yes = 6

NO = 4

1. Entropy of parent mode

$H = -0.6 \log_2(0.6) - 0.4 \log_2(0.4)$

$H = 0.971$ ✓

2. Info gain child nodes:

$A = a_1$: 4 Yes, 0 NO → Entropy = 0

$A = a_2$: 2 Yes, 4 NO → Entropy = 0.9183

$IG = 0.971 - [0.4(0) + 0.6(0.9183))$

$IG = 0.420$ ✓

3. Gini impurity Decreas

Parent : $1 - [0.6^2 + 0.4^2) = 0.48$

weighted child: $0.6(0) + 0.6(0.444) = 0.2667$ ✓

Decrease = 0.2133

Task 6 - cost complexity pouning

Given:

subtree errors = 3

subtree leaves = 4

single leaf errors = 8

$R_\alpha(T) = R(T) + \alpha |leaves|$

subtree cost: $3 + 4\alpha$

single leaf cost: $8 + \alpha$

At equal cost: $3 + 4\alpha = 8 + \alpha$

$3\alpha = 5$

$\alpha = 1.667$ ✓

for $\alpha = 0.01$:

subtree cost = 3.04

subtree leaf cost = 8.01

∴ dot not poune.

for $\alpha = 0.02$:

subtree cost = 3.08

single leaf cost = 8.02

∴ do not poune.

Task-7: Neural Networks and Back Propagation

Given $w_1 = 0.4$ $w_2 = 0.3$ $v = 0.5$ input = (1,2)

Target = 1 Learning rate = 0.5

1. Forward pass

$h = \sigma(0.4(1) + 0.3(2))$

$h = 0.7311$

Output: $\hat{y} = \sigma(0.5 \times 0.7311)$

$\hat{y} = 0.5904$

2. Cal Error

$E = \frac{1}{2}(1 - 0.5904)^2$

$E = 0.0839$

3. Backpropagation error terms

Output delta

$\delta_o = (1-\hat{y})(1-\hat{y})\hat{y}$

$\delta_o = 0.0991$

Hidden delta

$\delta_h = h(1-h)v\delta_o$

$\delta_h = 0.00974$

4. Gradients

→ Gradient of $v = -0.0724$

→ Gradient of $w_1 = -0.0097$

→ Gradient of $w_2 = -0.0194$

5. Updated weights

→ $v = 0.536$

→ $w_1 = 0.40$

→ $w_2 = 0.3097$

New loss:

$E_{New} = 0.0811$

Task 8: Support Vector Machine

given: $w = (0.5, 0.5)$

$b = -2$

a) verify support vectors

$f(3,3)$: $0.5(3) + 0.5(3) - 2 = 1$

$f(1,1)$: $0.5(1) + 0.5(1) - 2 = -1$

Margin width: $\frac{2}{\|w\|} = \frac{2}{\sqrt{0.5}} = 2.828$

b) classify points:

For (4,2):

$f(x) = 0.5(4) + 0.5(2) - 2 = 1$

class = +1 Distance = 1.414

for (1,2): $f(x) = 0.5(1) + 0.5(2) - 2 = -0.5$

class = -1

Distance = 0.707

c) Hinge loss

formula: $L = \max(0, 1 - y f(x))$

for (1,2) true label = -1:

$L = \max(0, 1 - 0.5) = 0.5$

for (2,2) true label = +1:

$L = \max(0, 1 - 0) = 1$

Task 9 Ensemble learning

Given:

→ Individual classifier accuracy = 0.70

→ classifiers are independent.

For 3 classifiers:

Majority voting accuracy

$$P = 3(0.7)^2(0.3) + (0.7)^3$$ ✓

78.4%

for 5 classifiers:

$$P = \sum_{k=3}^{5} \binom{5}{k} (0.7)^k (0.3)^{5-k}$$

83.692% ✓

Analysis:

→ Adding classifiers ↑ ensemble accuracy.

→ Majority voting reduces individual errors.

→ If errors are strongly correlated, ensemble improvements becomes smaller because classifiers make similar mistakes.

Task 10 - AdaBoost

Given: Total samples = 5

Initial weight = 1/5 = 0.2

Misclassified samples = 2 and 5.

1. weighted error

$$\epsilon = 0.2 + 0.2$$ ✓

$\epsilon = 0.4$

Task 11

2. learner weight.

$$\alpha = \frac{1}{2} \ln \frac{1-\epsilon}{\epsilon}$$

$$= \frac{1}{2} \ln \frac{0.6}{0.4}$$

$$\boxed{\alpha = 0.2027}$$

3. Un-normalised weights

→ Misclassified samples : 0.24495 each

→ correctly classified samples: 0.16330 each

Sample	weight
1	0.16330
2	0.24495
3	0.16330
4	0.16330
5	0.24495

4. Normalised weights

Sample	Net weight
1	0.1667
2	0.2500
3	0.1667
4	0.1667
5	0.2500

Analysis :

→ AdaBoost increase weights of misclassified samples.

→ correctly classified samples receive lower weights.

→ The next learner focuses more on difficult samples.

Task II - Discriminant functions.

Given: $\Sigma = \begin{bmatrix} 2 & 0.5 \\ 0.5 & 1 \end{bmatrix}$

$\mu_1 = (1,2) \quad \mu_2 = (3,1)$

a) cal discriminant direction

Formula: $w = \Sigma^{-1}(\mu_2 - \mu_1)$ ✓

inverse covariance matrix:

$$\Sigma^{-1} = \frac{1}{0.75}\begin{bmatrix} 1 & -0.5 \\ -0.5 & 2 \end{bmatrix}$$

Mean difference: $\mu_2 - \mu_1 = (2, -1)$

$\therefore w = (1.4286, -1.713)$

For equal priors, threshold: $w^T \frac{\mu_1 + \mu_2}{2}$

Threshold = 0.2857

b) classify $x = (2.5, 1) \Rightarrow$ $w^T x = 1.8571$ ✓

$\therefore$ 1.8571 is greater than 0.2857:

Class = ω_2

c) unequal priors

Given $P(\omega_1) = 0.8 \quad P(\omega_2) = 0.2$

New threshold: $0.2857 + \ln(4) = 1.672$ ✓

$\therefore$ 1.8571 is greater than 1.672.

Class = ω_2

Task +2 - classifier Evalation

a) accuracy, precision, Recall and f1-score

$$\text{Accuracy} = \frac{TP+TN}{\text{Total}} \qquad \text{Precision} = \frac{TP}{TP+FP} \qquad \text{Recall} = \frac{TP}{TP+FN} \qquad F1 = \frac{2PR}{P+R}$$

Metric	Model A	Model B	Model C
Accuracy	88%	80%	93%
Precision	43%	32%	66%
Recall	70%	85%	60%
F1-score	53%	46%	63%

b) Total expected cost

Formula: Cost = (FN × 50000) + (FP × 5000)

Model A: 30 (50K) + 90 (5K) → 1950000

Model B: 15 (50K) + 180 (5K) → 16,50,000

model C: 40 (50K) + 30 (5K) → 21,50,000

c). which model should be deployed?

→ Model c has the highest accuracy (93%)

→ Model c also has the highest F1-score (63%)

→ Model B has the lowest total cost [16,50,000).

→ Model B has highest recall (85%).